POWER BI

DATA SET – 3

NORTHWIND TRADERS

BUSINESS INSIGHT REPORT

By:

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1. INTRODUCTION:

Brief overview:

Northwind Traders is a sample trading company that buys and sells food and beverage products to customers across many countries. The dataset covers orders, products, product categories, customers, employees and freight (shipping) costs for the years 2013 to 2015.

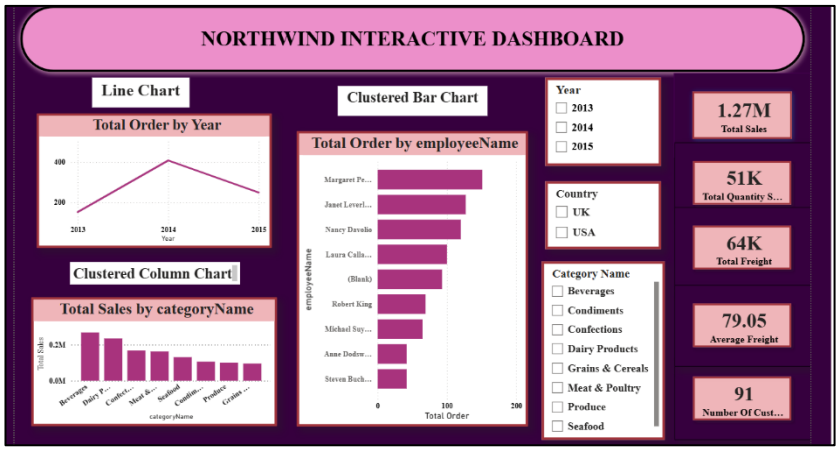
Purpose of the dashboard:

The Power BI dashboard turns the raw Northwind data into simple visuals and KPI cards so that sales, orders, customers, product categories and employee performance can be understood at a glance. Interactive filters let a user explore the data by year, country and category.

Purpose of the Business Insight Report:

This report explains what the dashboard shows, converts those observations into business insights, and gives practical recommendations that can help the company grow sales, plan inventory, control freight cost and target the right customers.

2. DASHBOARD OVERVIEW:



Purpose of the dashboard:

To give a single-page view of overall sales performance and allow the user to drill into any year, country or category.

Key components:

- **KPI cards**: Total Sales, Total Quantity Sold, Total Freight, Average Freight, Number of Customers
- **Line chart:** Total Order by Year
- **Clustered column chart:** Total Sales by Category Name
- **Clustered bar chart:** Total Order by Employee Name
- **Product summary table:** Product, Category, Total Sales, Quantity Sold
- **Clustered bar chart:** Number of Customers by Country

Interactive filters available:

- Year (2013, 2014, 2015)
- Country / Region (for example UK, USA)
- Category Name (Beverages, Condiments, Confections, Dairy Products, Grains & Cereals, Meat & Poultry, Produce, Seafood)
- KPI cards update automatically with every filter selection

3. KPI ANALYSIS:



KPI	VALUE	INTERPRETATION
Total Orders (TO)	809	The company handled 809 orders over the period. With 91 customers, each customer placed about 9 orders on average, which shows repeat buying.
Total Quantity Sold (TQS)	51,317 units	About 63 units per order, so orders are mostly bulk or wholesale in nature.
Total Freight (TF)	About 64K	Shipping cost is roughly 5% of total sales (1.27M), a manageable but noticeable cost.
Average Freight (AF)	79.05 per order	The typical shipping cost per order. Reducing this by even a small amount saves a lot across 809 orders.
No. of Customers	91	A small customer base spread over 21 countries, so each customer is valuable.
Total Sales	1.27M (1,265,793)	Average order value is about 1,565.

Key observation:

The business sells in large quantities to a small, loyal customer base, so keeping existing customers happy matters as much as finding new ones

4. PRODUCT CATEGORY PERFORMANCE ANALYSIS:

productName	categoryName	Total Sales	Total Quantity Sold
Côte de Blaye	Beverages	141,396.74	623
Thüringer Rostbratwurst	Meat & Poultry	80,368.67	746
Raclette Courdavault	Dairy Products	71,155.70	1496
Tarte au sucre	Confections	47,234.97	1083
Camembert Pierrot	Dairy Products	46,825.48	1577
Gnocchi di nonna Alice	Grains & Cereals	42,593.06	1263
Manjimup Dried Apples	Produce	41,819.65	886
Alice Mutton	Meat & Poultry	32,698.38	978
Carnarvon Tigers	Seafood	29,171.88	539
Rössle Sauerkraut	Produce	25,696.64	640
Mozzarella di Giovanni	Dairy Products	24,900.13	806
Ipoh Coffee	Beverages	23,526.70	580
Sir Rodney's Marmalade	Confections	22,563.36	313
Uncle Bob's Organic Dried Pears	Produce	22,044.30	763
Wimmers gute Semmelknödel	Grains & Cereals	21,957.97	740
Gudbrandsdalsost	Dairy Products	21,942.36	714
Ikura	Seafood	20,867.34	742
Perth Pasties	Meat & Poultry	20,574.17	722
Gumbär Gummibärchen	Confections	19,849.14	753
Flotemysost	Dairy Products	19,551.03	1057
Boston Crab Meat	Seafood	17,910.63	1103
Pâté chinois	Meat & Poultry	17,426.40	903
Pavlova	Confections	17,215.78	1158
Total		1,265,793.04	51317

CATEGORY	TOTAL SALES	SHARE OF SALES
Beverages	267,868	21.2%
Dairy Products	234,507	18.5%
Confections	167,357	13.2%
Meat & Poultry	163,022	12.9%
Seafood	131,262	10.4%
Condiments	106,047	8.4%
Produce	99,985	7.9%
Grains & Cereals	95,745	7.6%

High-selling categories:

Beverages (267,868) Dairy Products (234,507) and Confections (167,357). Together they contribute about 53% of total sales.

Low-selling categories:

Grains & Cereals (95,745), Produce (99,985) and Condiments (106,047).

Difference:

Beverages sold 172,124 more than Grains & Cereals, which is 179.77% higher.

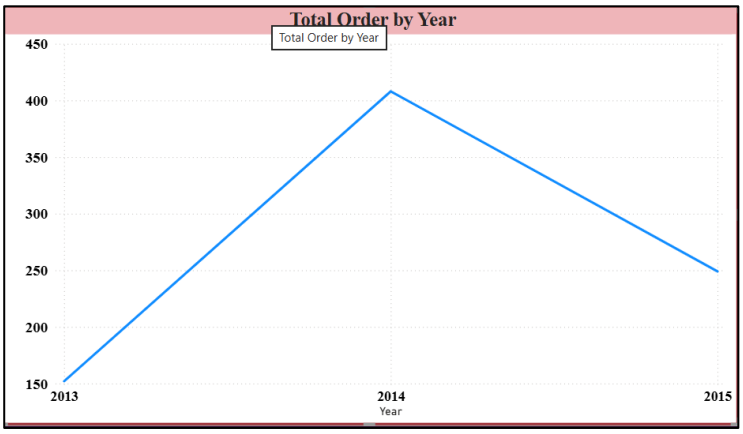
Observations:

- Sales range from 95,745 to 267,868 across the 8 categories.
- The top product, Côte de Blaye (Beverages), alone earned 141,397, which is about 11% of all sales.
- Meat & Poultry has only 6 products but earns 163,022 because its average price per unit (about 38.8) is the highest. Seafood has 12 products but a low price per unit (about 17.1), so its sales are lower.
- Only 36 of the 77 products generate 80% of total sales. The top 10 products give 44% of sales.

Key insight:

Revenue depends heavily on a few products and categories. High-price products matter more than high-volume products.

5. MONTHLY ORDER TREND ANALYSIS:



Year	Total Orders	Share of Orders
2013	152	18.8%
2014	408	50.4%
2015	249	30.8%

High:

2014 was the peak year with 408 orders, an increase of about 168% over 2013.

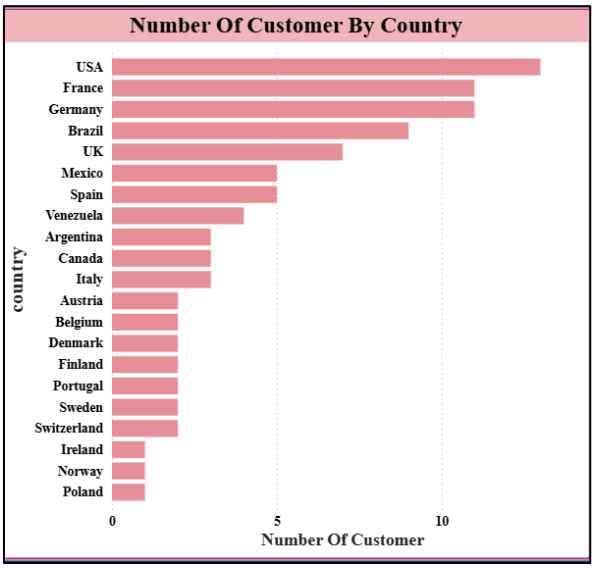
Low:

2013 had the lowest orders (152). 2015 fell by about 39% compared with 2014, although it is still 63.82% higher than 2013.

Insight:

The business grew strongly from 2013 to 2014, but order volume dropped in 2015. The 2015 data may cover only part of the year, so this should be confirmed in the dataset before concluding that demand has fallen. If it is a real drop, the company should investigate causes such as customer loss, stock shortages or seasonal effects.

6. REGIONAL ANALYSIS:





REGION	MAIN OBSERVATION	BUSINESS INSIGHT
USA	Highest number of customers (13, about 14%).	The USA is the largest customer market and should be protected with strong service and regular offers.
France and Germany	11 customers each (about 12% each). Together with the USA they hold 35 of 91 customers.	Europe is the strongest region overall: 54 of 91 customers (about 59%) are European.
Brazil, UK, Mexico, Spain	Brazil 9, UK 7, Mexico 5, Spain 5.	These are medium markets with room to grow.
Ireland, Norway, Poland	Only 1 customer each.	Very small presence, so either a growth opportunity or a market that is not worth heavy investment.

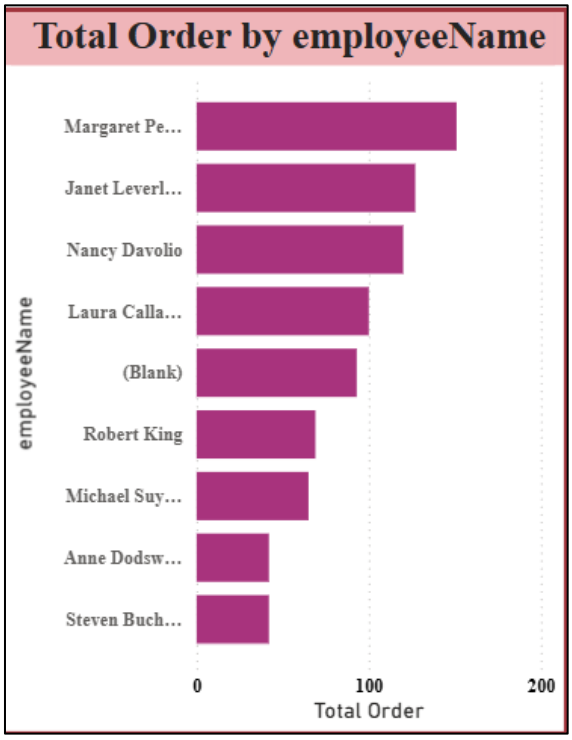
Overall:

The top 5 countries (USA, France, Germany, Brazil, UK) account for about 56% of all customers. The remaining 16 countries share the other 44%.

KPI change by region:

When the Country filter is applied (for example UK or USA), the KPI cards, category sales chart and order chart all update. This helps compare how each region performs on sales, orders and freight.

7.EMPLOYEE PERFORMANCE ANALYSIS:



EMPLOYEE	ORDERS	SHARE
Margaret Peacock	151	18.7%
Janet Leverling	127	15.7%
Nancy Davolio	120	14.8%

Laura Callahan	100	12.4%
(Blank)	93	11.5%
Robert King	69	8.5%
Michael Suyama	65	8.0%
Anne Dodsworth	42	5.2%
Steven Buchanan	42	5.2%

Low:

Anne Dodsworth and Steven Buchanan handled the fewest (42 each).

Difference:

The top performer handled about 3.6 times as many orders as the lowest performers. The top 3 employees together handled about 49% of all orders.

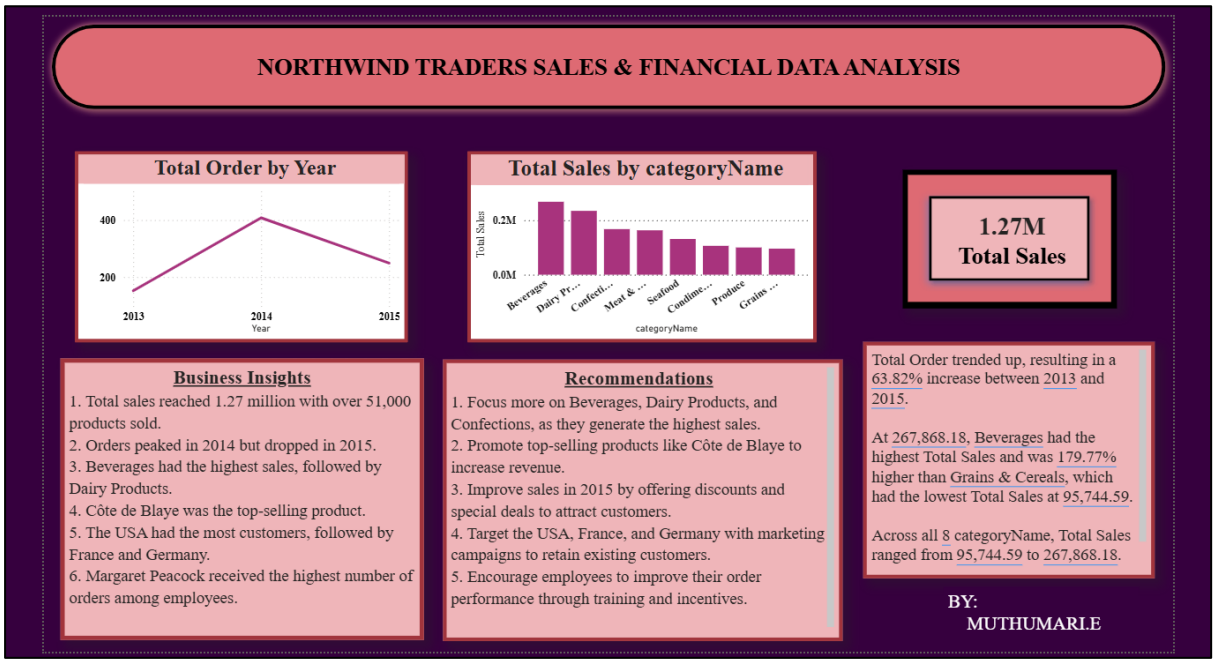
Change:

Workload is not evenly shared. Some employees carry far more orders than others.

Key insight:

A few employees drive nearly half of all orders. Also, 93 orders (11.5%) have no employee name recorded. This is a data quality issue that should be fixed so performance can be measured fairly.

7. BUSINESS INSIGHTS:



OBSERVATION (WHAT THE DATA SHOWS)	**INSIGHT** (WHAT IT MEANS FOR THE BUSINESS)
Total sales are 1.27M from 51,317 units across 809 orders.	The business has a healthy base of bulk, repeat orders.
Orders peaked at 408 in 2014 and fell to 249 in 2015.	Growth is not yet steady. Causes of the 2015 fall must be checked.
Beverages (267,868) lead and Dairy Products (234,507) follow.	These two categories carry the business and should get priority in stock and promotion.

Côte de Blaye alone earns about 11% of sales.	Heavy dependence on one product is a risk if its supply or demand changes.
The USA, France and Germany have the most customers.	Marketing effort will give the best return in these countries.
Margaret Peacock handled the most orders (151) while two employees handled only 42.	Workload and skills are uneven. Training and fair distribution can lift overall output.
Average freight is 79.05 per order, with about 64K total freight.	Shipping is about 5% of sales and offers savings if managed well.

9.BUSINESS RECOMMENDATIONS:

- **Product category strategy:** Focus stock and promotion on Beverages, Dairy Products and Confections. Push top products like Côte de Blaye, and either promote or drop low sellers like Chocolade.
- **Regional strategy:** Retain customers in the USA, France and Germany. Grow Brazil, UK, Mexico and Spain.
- **Inventory planning:** Keep high-volume items like Camembert Pierrot and Raclette Courdavault always in stock. Avoid overstocking slow items.
- **Monthly order planning:** Offer deals in slow months and prepare stock and staff before peak months. Find out why orders fell in 2015.

- **Freight management:** Cut the 79.05 average freight by negotiating shipping rates, combining regional orders, and setting a minimum order value for free shipping.
- **Customer targeting:** With only 91 customers, reward repeat buyers and follow up with those who ordered less in 2015.
- **Employee performance and data quality:** Balance the order workload, train the lower performers, and make employee name mandatory so there are no blank records.

10.CONCLUSION:

Northwind Traders achieved total sales of 1.27M from 809 orders and 51,317 units sold to 91 customers across 21 countries. Beverages and Dairy Products are the strongest categories, the USA, France and Germany are the largest customer markets, and Margaret Peacock handled the most orders. Sales depend heavily on a small set of products, categories, countries and employees, and orders grew sharply in 2014 before dropping in 2015, so protecting the strong areas while fixing the weak ones is the key to steady growth. The company should prioritise high-selling categories and products, target the top regions, plan inventory and monthly orders using trends, control freight cost, retain customers and balance employee workload. Power BI supports all of this by turning raw data into clear visuals and KPI cards, and its interactive filters for year, country and category allow quick drill-down without writing code, helping managers make faster data-driven decisions.